Channels to Targets: A Closed-Form Coordinate System for Silence-Blindness in LLM Observers

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Code: https://github.com/akhan531/Control-On-LLMs

Abstract

1 Introduction

An observer reads a transcript in which N analysts have each tested which candidate is responsible for a contamination event. From the record it is given, the observer must output the name of the responsible candidate and how confident it is in its answer. However, what the transcript shows of those tests varies, like if only k of the analysts file a report, while the remaining $N - k$ file nothing. The filings that never arrive are part of that record: whether the observer reads a missing report as a finding, as noise, or as nothing at all changes the belief it should hold, so an observer that conditions only on what appears can discard much of the evidence.

Our starting point is that this situation is small enough to solve exactly through Bayesian inference. Once the specific features of information, or channels, a transcript exposes are written down, the target posterior the observer should hold and the operations it must perform to reach that target both follow in closed form. An elicited answer is then not simply right or wrong: it has a location on an interval whose endpoints are reference beliefs, each the Bayesian posterior under a different reading of the transcript, and the distance and the direction are both measurable.

The design's asset is that the map from channels to targets is many to one: transcripts with different channels require the observer to perform different operations to absorb all the evidence available. However, if the target posterior is the same, a comparison across them holds the target fixed and varies only what the observer must do to reach it. This framework gives us a mathematical coordinate system to benchmark whether LLMs reason under the limits of their own knowledge, distinguishing what a transcript makes known from what its silences leave unknown rather than reading absence as no evidence at all.

- **An instrument.** A closed-form map from the channel set a transcript exposes to the operations required and the belief a correct observer should hold, with three reference beliefs and an exact pivot that makes three conditions share one target.
- **Localizing the direction of error.** Among the five configurations admitted by the competence gate, in the two conditions that do not enumerate the absences, under-weighting is the only direction of error observed, and a model comes to rest on the silence-blind belief more often when the absences must be recovered than when they are printed.
- **Separating identification from interpretation.** How much of a configuration's silence deficit is failure to notice the absences rather than failure to understand them is a per-configuration quantity the instrument measures, and enumerating the absences repairs the first without repairing the second.

- **The instrument's own limits, reported as results.** Two structural quantities are welded together by the geometry, one of our registered checks does not measure what it names, and our primary registered prediction did not hold.

2 The framework

Two candidates $A = \{a_0, a_1\}$ are under consideration and exactly one is responsible, with a uniform prior π. We take $|A| = 2$ throughout: the one-dimensionality that makes the word "coordinate" literal depends on it, and it is the source of a structural limitation we return to in Section 6. Each of the N analysts runs a binary assay returning POSITIVE or NEGATIVE. If a_0 is responsible, an assay returns POSITIVE with probability p; if a_1 is responsible, with probability $1 - p$; and the results are independent given which candidate is responsible. We take $1/2 < p < 1$, so a POSITIVE is more consistent with a_0 than with a_1 and tilts belief toward a_0. What the observer's transcript shows of these outcomes is fixed by a *disclosure rule*, and it is this rule that our conditions vary: some display every result, others only file the k POSITIVE reports and leave the $N - k$ NEGATIVE reports as absences to be recovered from N and the count of filed reports. In the dropout conditions, where only positives are filed, a positive additionally fails to be filed with probability δ. This is the bookbag-and-poker-chip paradigm of Phillips and Edwards [1966] conditioned on a disclosure rule. Across conditions, the transcript therefore displays different channels, as Table 1 sets out, and it is the channel set a transcript exposes that determines both the operations a correct observer must perform and the belief it should hold.

Write $L(a)$ for the probability that a given analyst's assay returns positive under candidate a, and let k be the number of positives among the N assays. Three beliefs are definable from the transcript alone:

$$b_{\text{pos}}(a) \;\propto\; L(a)^k, \tag{1}$$

$$q_\delta(a) \;\propto\; \big[(1-\delta)L(a)\big]^k \,\big[1-(1-\delta)L(a)\big]^{N-k}, \tag{2}$$

$$q_0(a) \;\propto\; L(a)^k \,\big(1-L(a)\big)^{N-k}. \tag{3}$$

Equation (1) reads the positive filed reports as if they were the whole record. Equation (2) conditions on presences and absences alike, discounting each absence by the chance it reflects a dropout rather than a negative finding. Equation (3) treats every absence as certain disconfirmation.

We write $b^\star := q_0$ at $\delta = 0$ only, where it is the correct posterior. At $\delta > 0$ the correct posterior is q_δ and q_0 becomes the *credulous point*, in that it takes every absence at face value as a negative, lying past correct in the direction of over-reading by the strict ordering of Lemma 1, so draws below correct in the dropout conditions are readable rather than merely wrong.

Proposition 1 (The pivot)**.** *At $\delta = 0$, $q_\delta = q_0 = b^\star$.*

Proof. Setting $\delta = 0$ in (2) makes the discount factor $(1-\delta)$ vanish from both exponentiated terms, so the silence factor $\big[1-(1-\delta)L(a)\big]^{N-k}$ becomes $\big[1-L(a)\big]^{N-k}$, which is exactly the negative-result factor of (3). The two expressions are identical term by term. □

The consequence is the design's asset: conditions that print the absences, enumerate them, or leave them to be derived all return the same target, so their comparison is controlled. Every reference belief in this paper is computed over exact rationals, and Proposition 1 is asserted as exact rational equality across all fourteen cells rather than to a tolerance; no floating point appears anywhere in the model.

Lemma 1 (Ordering). Let $|A| = 2$ and $1/2 < p < 1$, and consider a transcript with at least one absence ($N - k \geq 1$). For every $\delta \in (0, 1)$ the beliefs b_{pos}, q_δ, q_0 are strictly ordered in log-odds, and q_δ moves monotonically toward b_{pos} as δ increases.

Proof. Let a_0 be the candidate with $L(a_0) = p$ and a_1 the other, so $L(a_1) = 1 - p$ and $p > 1/2$ gives $L(a_0) > L(a_1)$. Write $\Lambda(b) := \log \frac{b(a_0)}{b(a_1)}$ for the log-odds a belief b places on a_0. The three beliefs share the presence term $k \log \frac{p}{1-p}$ and differ only in how each of the $N - k$ absences is

Table 1: The map. Each condition's transcript exposes a different channel set, which determines both the operations a correct observer must perform and the belief it should hold. Rows 2 through 4 differ in operations and agree on target, by Proposition 1.

condition	channels in transcript	operations required	target
PARTIAL_BLIND	positives	aggregate	b_{pos}
FULL	positives, printed negatives	aggregate	$b^\star$
ARITH_D0	positives, enumerated absences	convert, aggregate	$b^\star$
PARTIAL_RULED	positives, derivable absences	identify, convert, aggregate	$b^\star$
ARITH_D30	positives, enumerated absences, δ	convert, aggregate, discount	q_δ
RULED_D30	positives, derivable absences, δ	identify, convert, aggregate, discount	q_δ

scored:

$$\Lambda(b_{\text{pos}}) = k \log \tfrac{p}{1-p}, \qquad \Lambda(q_\delta) = k \log \tfrac{p}{1-p} + (N-k)\, g(\delta),$$

$$\Lambda(q_0) = k \log \tfrac{p}{1-p} + (N-k)\, g(0), \qquad g(\delta) := \log \frac{1-(1-\delta)p}{1-(1-\delta)(1-p)}.$$

Because $p > 1-p$, the numerator of g is the smaller factor, so $g(\delta) < 0$ for all $\delta \in [0,1)$: every absence strictly lowers the log-odds on a_0. Moreover

$$g'(\delta) = \frac{p}{1-(1-\delta)p} - \frac{1-p}{1-(1-\delta)(1-p)} > 0,$$

since $x \mapsto x/(1-(1-\delta)x)$ is strictly increasing and $p > 1-p$, so g climbs strictly from $g(0) = \log \frac{1-p}{p} < 0$ to $g(1) = 0$. Hence for any transcript with an absence ($N-k \geq 1$) and $0 < \delta < 1$,

$$\Lambda(q_0) < \Lambda(q_\delta) < \Lambda(b_{\text{pos}}),$$

the claimed strict ordering, and $\Lambda(q_\delta)$ rises monotonically to $\Lambda(b_{\text{pos}})$ as δ increases. The endpoints recover Proposition 1 at $\delta = 0$ and b_{pos} as $\delta \to 1$. □

Four operations appear in Table 1. To *identify* is to determine which reports are absent; once N is stated this is a single subtraction, so noticing that reports are missing and naming which ones are missing are not separable stages. To *convert* is to read an absence as a negative finding. To *aggregate* is to combine the findings into a posterior. To *discount* is to weigh each absence against the stated dropout rate. The three conditions sharing a target require the strictly nested sets $\{a\} \subset \{c, a\} \subset \{i, c, a\}$, which is the ladder Example 2 walks. Two honest limits follow: aggregation is never identified in isolation, because no condition in our design sits below $\{a\}$, and discounting is identified only by the $\delta = 0.3$ conditions.

3 The instrument

The framework of Section 2 is a claim about any observer that reads such a transcript. We now put real LLM observers on it, running the conditions of Table 1 as prompts and reading each model's answer as a location on the same coordinate. The seven configurations run four model families: `openai/gpt-5.6-sol`, `z-ai/glm-5.2`, `deepseek/deepseek-v4-flash-0731`, and `meta-llama/llama-3.3-70b-instruct`. Every family except the Llama one contributes two configurations that differ in reasoning setting and the paired token budget, which is what the "-high" suffix on a configuration name denotes (abbreviated "-hi" in Figure 2); Appendix C gives the full table.

3.1 Realizing the coordinate system for LLM observers

Answers are elicited as two enumerated fields, a candidate and a two-level confidence, giving four ordinal levels. We elicit an ordinal level rather than a raw probability on purpose: a numerical posterior would confound the ability to reason about the evidence with the ability to carry out the arithmetic of Bayesian updating, and coarsening the answer to four ordered levels strips out the noise of that arithmetic and keeps the signal we mean to measure, whether the observer moves its belief in

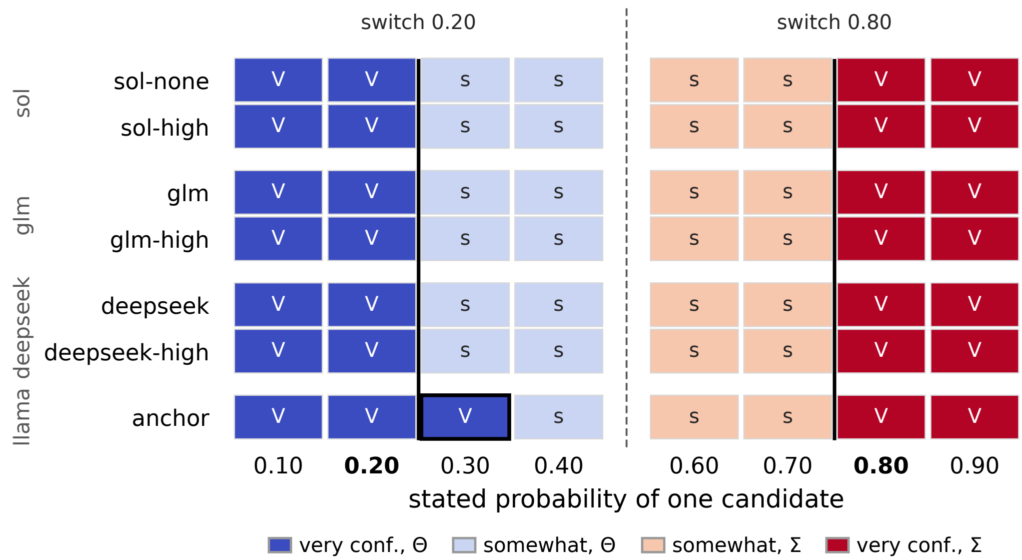

Figure 1: The calibration confidence map. Each row is one configuration; reading its cells outward from even odds, the answer switches from somewhat to very confident. Six of the seven switch together at the 0.20 and 0.80 ticks (bold); `anchor` is the exception, holding very confidence one step further in, to 0.30 on the low side (boxed). Colour encodes the answer given most often in each cell: hue is the chosen candidate, Σ or Θ, and shade its confidence, dark for very and pale for somewhat.

the right direction and far enough. The probability ranges these four levels occupy were measured, not assumed. In a calibration pre-test, each configuration is handed a stated probability for one candidate directly, with no inference to perform, and answers in the same two enumerated fields. Sweeping that stated probability (32 stimuli, three draws each, across all seven configurations, 659 of 672 answers usable) and taking, at each step, the answer a configuration gave most often across its draws, we read off where on the scale a model switches from very confident to only somewhat confident. That switch sits at 0.20 on the low side and 0.80 on the high side (Figure 1): a model is very confident in a candidate only once its probability clears 0.80, very confident in the other candidate below 0.20, and merely somewhat confident closer to even odds. We located the low edge and the high edge separately rather than assuming the low side mirrors the high, and for six of the seven configurations they came out symmetric about the 0.5 midpoint; only `anchor` differed, by 0.1. One caveat on precision: the probe sampled probabilities in steps of 0.10, so it cannot tell the measured edge at 0.20 apart from a nominal cut at 0.25, both falling in the same step, which means the level assignment agreeing under either cut reflects the coarseness of the probe rather than a demonstrated robustness. Bands were never shown to the models.

Four constraints are enforced at build time. Every cell asserts that the silence-blind belief and the target fall in different levels, which holds fourteen for fourteen at $\delta = 0$ and nine for nine at $\delta = 0.3$. A build-time forbidden-word list bans silence vocabulary together with Bayesian and computational vocabulary, so no prompt cues the operation it is testing for. Every stimulus appears in both label orderings. Count-control cells invert the correct level, so a configuration that always answers high-confidence is separable from a competent one. Appendix B reproduces the prompts, the response schema, and one rendered stimulus per condition.

The four levels sit in measured bands with deliberate gaps between them: level 0 is $[0, 0.20]$, level 1 is $[0.40, 0.45]$, level 2 is $[0.55, 0.70]$, and level 3 is $[0.80, 1]$, so the ranges 0.20–0.40, 0.45–0.55, and 0.70–0.80 fall under no level at all, and the viability filter is how the design survives those gaps rather than ignoring them. The gaps are kept on purpose: they are the transition zones where a model's answer is not reliably one level, so extending the borders to close them would score answers where the level is ambiguous, turning threshold noise into apparent signal. A cell is usable only if the belief its correct answer should express lands inside a band, giving that answer a definite level; a cell fails on gap placement when that belief lands in one of the empty ranges instead, where the scale offers no level to score against. Five of the fourteen cells fail this way on their dropout target: two put it at 0.366 and 0.392, inside the 0.20–0.40 gap between bands 0 and 1, and three put it between 0.512 and 0.524, inside the 0.45–0.55 gap that straddles the prior, so in each the correct belief has no level and the cell is dropped. That is what reduces fourteen cells to the nine carrying the dropout conditions. The filter is also stricter than cell validity requires, since a cell carrying those conditions needs its dropout target to be distinguishable from both the silence-blind belief and the credulous point, which is why stating a dropout rate can change the correct answer.

3.2 The competence gate

Before reading any model against the coordinate, we check that it can read the record at all. A competence gate separates the configurations that handle the plainly stated version of the task from those that do not, so that a later failure on the harder conditions can be read as silence-blindness rather than as an inability to do the arithmetic. The gate turns on two conditions built from the same underlying record. In the printed-negatives condition (FULL), every result, positive and negative, is displayed and completeness is asserted, so reaching the correct answer is a matter of aggregation; in the derivable-absences condition (PARTIAL_RULED), only the positive reports appear, and the negatives must be inferred from the stated analyst count and filing rule. A configuration that fails the printed-negatives condition is excluded, because a failure on the derivable-absences condition from a configuration that also fails the printed one is a floor effect rather than a silence result.

To measure where a model lands on the coordinate, we need a distance in levels. We now define the statistics that report it. We report $s_1 = |\ell_{\text{obs}} - \ell_{\text{cor}}|$, the signed $s_2 = \ell_{\text{obs}} - \ell_{\text{cor}}$ with positive values meaning under-weighted toward the target, and $s_3 = P(\ell_{\text{obs}} = \ell_{\text{cor}})$, where ℓ_{obs} is the level of observer reports and ℓ_{cor} is the level of the correct target read off Table 1.

Five of seven configurations are admitted, at mean s_1 of 0.200, 0.421, 0.457, 0.752 and 0.764 bins; the two excluded sit at 1.609 (`deepseek`) and 1.800 (`anchor`). The gap between 0.764 and 1.609 means membership does not depend on where in that interval the bar is placed.

The configuration the gate excludes at 1.800 is also the one that fails the pre-test's reachability check, the only one whose switch point moves between wordings, and the only one with nonzero low-high asymmetry, so four instrument checks and one behavioural gate select the same configuration. Across five experimental arms the campaign holds 7,051 usable draws of 7,210 attempted, at five draws per stimulus; the pre-test described above contributes a further 659 of 672 on a separate stimulus set at three draws per stimulus. Predictions, gates, and kill conditions were fixed in a specification document and implemented as executable decision rules; Section 6 scores them and states plainly what our record does and does not establish about when they were fixed.

4 Example 1: which direction does the error run?

Whether language models update too little or too much is contested, and the two closest anchors disagree on populations that do not overlap. Imran et al. [2025] report that pre-trained-only checkpoints update less than Bayes requires, with the shortfall shrinking as models scale, measured over token log-probabilities; they describe the shortfall itself as unexplained and note that it may reflect how improbable their evidence strings are, so we take the direction from them and not a mechanism. Samanta et al. [2026] evaluate instruction-tuned models across turns and report both directions, with the tendency shifting across scale: smaller models stay near the middle of the interval, while larger ones can push toward its extremes when the evidence is skewed. Under-use relative to Bayes has been called conservatism since Edwards [1968]. Imran et al.'s expected update is itself elicited from the model, read off its own token probabilities, and where a closed-form reference is available elsewhere, it is computed from evidence the observer can see. Gupta et al. [2025] show that updates track the Bayesian posterior closely when enough of that evidence has accumulated in the model's context, with residual deviation traceable to a model's own miscalibrated prior rather than to the update itself. Our target is computed from the disclosure rule over exact rationals, so the question here is not whether a belief moves correctly on what the transcript displays but whether it moves at all on what it withholds.

Both anchors were selected after our data existed: the campaign's predictions were registered, the literature engagement was not.

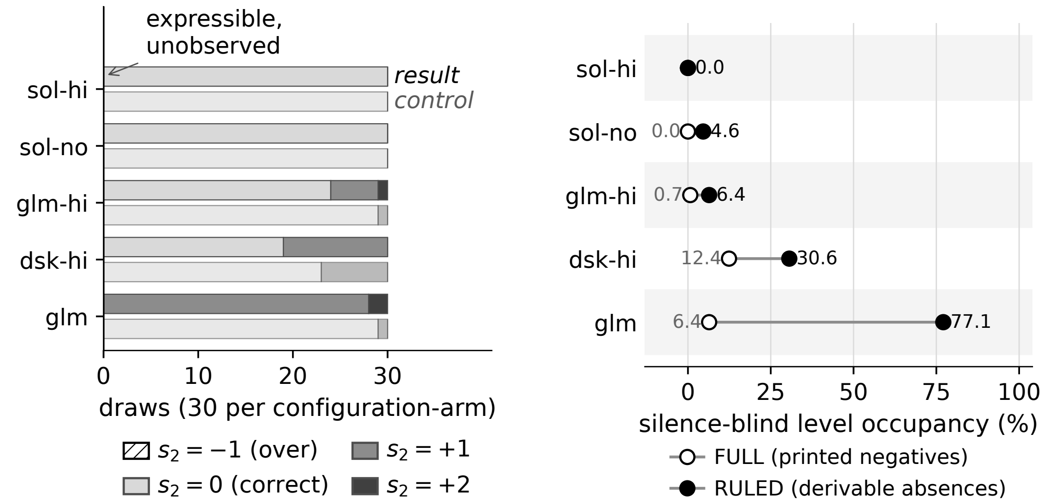

Figure 2: **(a)** Signed displacement s_2 on the three cells where over-weighting is expressible, five admitted configurations, 150 attempted draws per arm; the result arm is derivable-absences, the control (gate-selected) is printed-negatives. The $s_2 = -1$ (over-weighted) segment has zero width in every bar, which is the finding. **(b)** Occupancy of the silence-blind level, the percent of usable draws landing exactly on it, under the printed-negatives condition (open marker) and the derivable-absences one (filled), for the five admitted configurations over all 14 cells at $\delta = 0$. Occupancy rises in four of the five and falls in none, reaching 77.1 percent for `glm`; `sol-hi` sits at zero in both. Configurations are abbreviated (`dsk`=deepseek).

The comparison uses two channel sets, printed negatives against derivable absences, which require different operations and share a target by Proposition 1. Because the target lies below the silence-blind belief in every cell, an answer that under-weights the silence stays high, above the target, while one that over-weights it falls below. The first eleven cells have a target that already rests on the bottom level (Appendix A), so nothing lies below it and over-weighting is structurally inexpressible. We therefore built three cells, D1–D3, with the target one level up ($\ell_{\text{cor}} = 1$ rather than 0), specifically so that an answer can fall on either side of it; these three are where under-weighting and over-weighting are both expressible, and the direction-of-error result is measured on them by design.

On those three cells, the derivable-absences condition runs 47 to 0 toward under-weighting across the five admitted configurations (Figure 2a), on 150 attempted draws of which 150 were usable. The competence gate admits configurations by their performance on the printed-negatives condition, not on this derivable-absences one, so the arm that carries the result played no part in the selection. At the registered unit of analysis, which reduces one configuration's several runs of a single cell (its repeated samples and the two mirrored, label-swapped presentations of that cell) to the single ordinal level they most often returned, eight of the fifteen configuration-by-cell units are displaced toward under-weighting and none toward over-weighting (two-sided exact sign test, $p = 0.008$); the remaining seven carry no displaced mass and are undefined rather than failed. This direction persists in the printed-negatives condition, 9 to 0 on the same 150 attempted draws, five of fifteen units displaced, $p = 0.063$.

We also ask how often belief comes to rest exactly on the silence-blind level, and whether that changes with the channel. Occupancy of that level, the fraction of usable draws that land on it, is measured for the five admitted configurations over all cells minus the admission exclusions, under both the printed-negatives condition and the derivable-absences one at $\delta = 0$ (Figure 2b). That level is one of the two above the prior: the uniform prior at 0.5 falls inside the band gap that straddles it exactly, so two levels lie wholly below it and two wholly above, with none ambiguous. Occupancy rises when the negatives must be inferred rather than read, and falls for none of the five, so a model settles on the silence-blind belief more readily precisely where the silence has to be recovered.

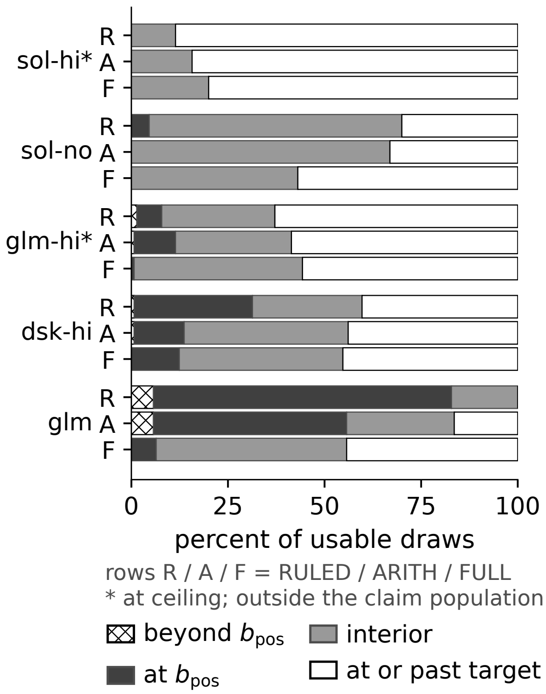

Figure 3: Four-way mass split (beyond b_{pos}, at b_{pos}, interior, at or past target) across the rungs RULED/ARITH/FULL, all cells minus admission exclusions, denominators varying by configuration. Starred configurations are at ceiling, outside the claim population (`dsk`=deepseek).

An ordinal extremity account predicts the opposite of what we see: on a four-level scale, the two middle levels are the ones an extremity effect would evacuate outward, so it predicts mass accumulating at the two endpoints. Of the 47 displaced draws, 44 land at the level of the silence-blind belief, which is not an endpoint, 3 land at the top level, and none at the bottom, so total endpoint mass is 3 in 150 attempted draws. The silence-blind account names an interior level in advance; extremity names the two levels that stay nearly empty. In all three configurations with displaced mass to test, mass at the silence-blind level outnumbers mass at the endpoint.

One-sidedness here is a property of these two conditions and not of the instrument. The over-direction is recoverable with the same models once the channel set changes, as Section 5 shows at both dropout rates, so the objection that our roster is too small to produce it becomes the observation that our instrument sees it only where the transcript enumerates the absences, which is the thesis.

This example required no new inference calls and runs entirely on draws collected before it was conceived.

5 Example 2: identification or interpretation?

When a system fails to use the absences, it may be failing to notice them or failing to understand what they mean in context. In general, these two are confounded, because a task that makes the absences easier to see usually also makes them easier to interpret.

Table 1 separates them without a new task. Its three shared-target conditions (FULL, ARITH_D0, and PARTIAL_RULED) have strictly nested operation sets, so moving up the ladder deletes exactly one operation and changes nothing else, and the deletion is performed by the transcript rather than by an instruction. This does not have to be designed in; it falls out of the map.

The added rung contributes 980 calls, 959 usable, over a stimulus set fixed at a content hash; Section 6 states what that hash does and does not establish.

Deleting the identification step frees the answers. Each pair below runs from the derivable-absences condition to the enumerated one, the direction in which reaching the target posterior takes one operation fewer; read that way, the more operations the target requires, the more mass sits at the silence-blind belief. Residence there collapses in every non-ceiling admitted configuration once the step is removed, from 4.6 to 0.0, from 30.6 to 12.9, and from 77.1 to 50.0 percent of usable draws over all cells minus the admission exclusions (Figure 3): the absences were going unnoticed. But the freed answers do not arrive at the target. Interior mass, the mass strictly between the silence-blind belief and the correct one, rises in every one of them as that step comes out, from 65.4 to 66.9, 28.4 to 42.4, and 17.1 to 27.9. This pooled rise is not uniform across cell geometries: for sol-none it comes entirely from the distance-3 cells, while the distance-2 cells stay flat at 70.0 percent. Identification is repaired and conversion is not. Each half names a different comparison: identification is the operation the derivable-absences condition demands and the enumerated one drops, so its repair is the collapse above, read from the former to the latter; conversion is the operation that still separates the enumerated condition from the one that prints every result and asks only for aggregation, so its failure is why the freed mass rises into the interior rather than reaching the target, even with identification handed over. One further configuration, flagged at ceiling by the registered guard and therefore outside this population, moves the other way, from 6.4 to 10.7, and we report it.

The split is a per-configuration quantity and not a constant. All five admitted configurations receive a well-defined verdict: three numeric positions and two adjudicated at ceiling by the registered guard. Writing λ for the share of the gap between the two conditions that enumeration leaves unclosed, the closures span 28 to 114 percent of that gap, with the top of the range falling past the printed-negatives floor. The registration for this rung carried no numeric bounds, so we report λ as a measured descriptive quantity rather than scoring it against a prediction.

Once a dropout rate is stated the same handout does not merely fail to help; for three of the seven configurations it hurts, moving them 0.034, 0.100 and 0.232 bins further from the target. At $\delta = 0.3$, no configuration's distance to the target falls by more than 0.125 bins when the absences are enumerated rather than left to be inferred at that same dropout rate, and a worst-case reassignment of unusable draws moves that bound to 0.20. That covers all seven configurations and needs no gating argument. It is also where the geometry does work: the target moves to the discounted belief and the undiscounted one becomes a credulous point past correct, so draws below correct are readable as over-reading rather than as noise. One contrast we had fixed at a checkpoint failed on this arm and is retired rather than rescued.

The obvious alternative is that enumeration changes the record's length rather than its content, and there is real evidence for it: the count-control concordance fails for three configurations, and both ceiling configurations move the wrong way under enumeration, by 0.043 and 0.114 bins, which is the shape a length cost would take. Two things stand against it. Length cannot account for a gain of 0.493 bins, and a per-line effect fails to appear anywhere, ranging from -0.89 to $+0.17$ after a collinearity audit bounding the relevant correlation at 0.34. The concordance test also inherits the defect we describe in Section 6: its constancy criterion is confounded in the count-control group and clean in the group carrying this example's claims.

The two halves meet here. Once the absences are enumerated, over-reading appears without any dropout rate at all: two over-reads in 150 expressible admitted draws on the enumerated condition, against none in 300 on the two conditions that do not enumerate, and the two configurations producing them are exactly the two with the largest repairs. The direction of error flips with the channel set, at both dropout rates.

This recovers, by a different method, the failure Min et al. [2026] report as over-closure: treating missing support as settled is the closed-world assumption applied where it does not hold, which in our coordinates is the credulous point. They ask whether an absence licenses a negative answer at all, holding the question fixed and varying how coverage is expressed; we assume licensing and ask how far the belief should move, holding the target fixed and varying which operations the transcript requires.

6 What the instrument cannot identify

Our specification defined a three-band outcome space in advance: a pass required the effect in at least two of three families, zero families counted as falsification with the registered consequence

Table 2: Registration scoring across three tiers. Tier labels record where an item was registered, not that its ordering against the campaign was verified; see Section 6.

item	tier and where registered	outcome
P3 (primary): gap > 0.5 in ≥ 2 of 3 families	Tier 1, specification, internal timestamp	Falsified under a strict reading of our own P5 exclusion clause (0 of 3); held in 1 of 3 under a loose one. The single family that clears it at 0.821 also fails P5. Both readings reported.
P1 and P2, the two halves	Tier 1, same	1 of 3 each, in *different* families. No family carries both halves.
Non-control cell bar	Tier 1, amended	Specification text says 8 of 11; executed as 6 of 8. The non-control set has eight members, so the original figure was a miscount. Proportion $0.727 \to 0.75$. Dated amendment.
Enumerated-rung λ predictions	Tier 2, content hash	Directional, no numeric bounds. Scored as directional.
Enumerated-rung ceiling guard	Tier 2, same	Sharp 0.15-bin threshold, scored cleanly.
Cross-regime contrast	Tier 3, fixed at a checkpoint, absent from the specification	Did not hold. Retired rather than rescued.

that this becomes a negative-results report, and one family was neither. Naming the bands is what makes the result below reportable rather than merely disappointing. One thing our record cannot do is order itself. Nothing in the repository establishes that any registration predates the calls it registers, for the specification, the stimulus set, or the band pre-test alike; version control begins after both campaigns, draw records carry no wall-clock field, and the specification file was edited after the campaign it registers. What is verifiable is content integrity and deterministic regeneration, and we describe every decision rule as fixed at a checkpoint before its test ran, which is a claim about our process and not about verified ordering.

The instrument's limits are not a list of unrelated caveats. Three of them are the same shape: a pair of properties one would want to vary independently, fused by the geometry of the construction.

First, count and weight are anti-correlated by construction: holding the target fixed while raising the number of absences from four to twelve forces the assay sensitivity down from 0.90 to 0.56, and the evidence the silences carry falls from 5.61 through 1.83 to 0.71 nats, so tripling the count drops the evidence roughly eightfold. This is closed form and needs no draws. Second, over-reading is testable only in the weak-evidence corner, and the coupling is forced: the three cells where it is expressible carry silence evidence between roughly fifty and sixty times weaker than the strongest cells in the grid, because over-reading needs headroom below the target, headroom requires the target to be high, and a high target means the silences carried little. Third, the response scale makes over-reads harder to see than under-reads: in those same cells the target sits inside the narrowest band in the instrument, with a gap of 0.20 below it and 0.10 above, so registering an over-read requires roughly twice the excursion in probability that registering an under-read does. That asymmetry ships with the zero reported in Section 4.

A fourth limit is our own. One registered check requires the modal answer to stay constant as the report count varies, but in the count-control group the silence-blind belief is pinned while the correct level moves, so a configuration tracking the target fails the check and one parked at the silence-blind belief passes it. The most silence-blind configuration in our roster returns a perfectly constant row. This is why Table 2 reports the primary outcome under both readings of that check's exclusion clause.

A fifth is that our normalised statistic is span-dependent, since the spans available across the control sweep differ and cannot be pooled; we retire it and report it rather than dropping it quietly. Its companion is structural and is a fourth instance of the pattern above. With two candidates the silence-blind belief is at least 0.5 by construction, so it occupies only two of the four bins and the falsification test is always a one-bin discrimination; in the cells carrying Section 4's result the intermediate region is empty outright, so the fair symmetry test and the instrument's narrowest limitation are the same geometry. The repair is stated with the limitation: both follow from fixing the candidate set at two, an axis we dropped deliberately and mark for revisit.

A sixth is that the dropout conditions are mechanically easier under an ordinal statistic and cannot serve as the harder test they appear to be. Relatedly, the one-sidedness reported in Section 4 belongs to the two conditions that do not enumerate the absences and not to the no-dropout regime as such: enumeration produces over-reading with no dropout at all, and under stated dropout 39 of 620 draws (6.3 percent) on the derivable-absences arm and 91 of 623 (14.6 percent) on the enumerated arm, across all seven configurations, sit below correct.

7 Related work and limitations

Our correct update censors the record where the silence-blind belief trims it, in the sense of de Punder et al. [2026], and the mass censoring restores is exactly the silence probability: their censored log score adds to the trimmed one a term in the probability of the omitted region, which is our factor $(1 - L(a))^{N-k}$. That the trimmed quantity forfeits non-negativity is established in de Punder et al. [2024] rather than here. The same distinction appears in the knowledge-base literature as the gap between closed-world and open-world readings of a missing fact [Razniewski et al., 2024]. Our contribution is the instrument built on that distinction, not the distinction.

That language models mishandle absent evidence is known [Min et al., 2026, Fu et al., 2025]. Those results ask whether an absence licenses an inference at all; we ask how far a belief should move once it does, and only the second question requires a target computed from the disclosure rule.

Belief dynamics in multi-agent settings already admit martingale treatments [Choi et al., 2025]. The dynamical frame is not our contribution here. Our setting is single-round and static by choice, because holding the target fixed is what makes the channel-set manipulation a controlled comparison.

Three limitations bound what we claim: the roster is seven configurations across four model families, three open-weight and one closed, which is no broader than the rosters our two closest anchors use; the candidate set is fixed at two, which is what forces the one-bin degeneracy described in Section 6; and the setting is single-round and static.

Reproduction. Stimuli, decision rules, draw records, and analysis scripts are available at `https://github.com/akhan531/Control-On-LLMs`, commit `9ade4d4`, together with a provenance statement describing exactly what the record establishes and what it does not.

Acknowledgments

This work was supported by the Rutgers Summer Undergraduate Research Fellowship (SURF) program. The author thanks Dr. Laurent Burlion for his guidance and supervision.

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Table 3: The fourteen cells. N, k, and p are inputs; the remaining columns are computed from them. b_{pos} is the belief that ignores the absences and q_0 the correct posterior at $\delta = 0$, with ℓ_{pos} and ℓ_{cor} their ordinal levels. Beliefs are exact rationals, shown to three places.

cell	N	k	$N-k$	p	b_{pos}	q_0	ℓ_{pos}	ℓ_{cor}
A1	5	2	3	0.81	0.948	0.190	3	0
A2	8	3	5	0.67	0.893	0.195	3	0
A3	10	4	6	0.67	0.944	0.195	3	0
A4	13	5	8	0.62	0.920	0.187	3	0
A5	14	5	9	0.67	0.972	0.056	3	0
B1	11	3	8	0.57	0.700	0.196	2	0
B2	5	1	4	0.62	0.620	0.187	2	0
B3	8	2	6	0.59	0.674	0.189	2	0
C1	6	2	4	0.88	0.982	0.018	3	0
C2	8	3	5	0.79	0.982	0.066	3	0
C3	11	4	7	0.73	0.982	0.048	3	0
D1	5	2	3	0.56	0.618	0.440	2	1
D2	8	3	5	0.54	0.618	0.421	2	1
D3	10	4	6	0.53	0.618	0.440	2	1

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A The cell design

Each row of Table 3 is a *cell*: one scenario put to the observers, fixed by an analyst count N, the number of filed positives k, and a per-assay strength p. The conditions of Table 1 are run on each of these fourteen scenarios, and the draws they return are the data behind every result in the paper. In the table N, k, and p are the inputs; every other column is computed from them: the absences $N-k$, the silence-blind belief b_{pos} and the correct posterior q_0 at $\delta = 0$, and the levels ℓ_{pos} and ℓ_{cor} that those two beliefs occupy.

The cells fall into four blocks. **A** is the span-three backbone: the silence-blind belief sits at the top level ($\ell_{\text{pos}} = 3$) while the correct answer sits at the bottom ($\ell_{\text{cor}} = 0$), the widest displacement the scale allows, with N and k varied so that a level-three blind belief is built from different amounts of evidence. **B** is the span-two backbone, the same construction one level before, with the blind belief at level two; together A and B measure under-weighting at the two levels the blind belief can occupy. **C** is a count control at level three: b_{pos} is held essentially constant (0.982 across the block) while k varies, so a configuration that reasons about the evidence answers the same across C1–C3 whereas one that merely tracks the number of filed positives does not, which separates competence from count-following. **D** is a count control at level two, and the one block built with $\ell_{\mathrm{cor}} = 1$ by design: holding b_{pos} fixed at 0.618 plays the same control role as C, but placing the correct answer one level off the floor is what makes over-weighting expressible. D is the only block where an answer can land above the target as well as below it, and we built it that way on purpose so that under-weighting can be tested against over-weighting rather than only measured.

B Prompts and elicitation

There is no system prompt. Each draw is a single user message carrying the entire prompt below; no separate system or developer message is sent. The model returns a structured object validated against this schema, with strict decoding and no partial credit:

```
{"type": "json_schema",
 "json_schema": {"name": "silence_probe_v2_ordinal", "strict": true,
   "schema": {"type": "object",
     "properties": {
       "candidate":  {"type": "string", "enum": ["SIGMA", "THETA"]},
       "confidence": {"type": "string",
                      "enum": ["Somewhat confident", "Very confident"]}},
     "required": ["candidate", "confidence"],
     "additionalProperties": false}}}
```

The two enumerated answer fields are therefore `candidate`, allowed values `SIGMA` and `THETA`, and `confidence`, allowed values `Somewhat confident` and `Very confident`. Together they give the four ordinal levels of Section 3.

The six conditions of Table 1 are generated by three scripts: `silence_v2_stimuli.py` (FULL, PARTIAL_BLIND, PARTIAL_RULED, and the banked RULED_D30), `arith_d0_stimuli.py` (ARITH_D0), and `arith_d30_stimuli.py` (ARITH_D30). All three import and assert the same forbidden-word list, so the no-silence-vocabulary property holds across every condition, not a subset. The list has nineteen terms:

```
silent, silence, absent, absence, missing, did not report, no report,
nothing was filed, unreported, withheld, omitted, posterior, bayes,
prior probability, likelihood ratio, probability that, update your,
calculate, compute
```

Below is one fully rendered stimulus per condition, all from cell `A1` ($N = 5$, $k = 2$, positive rates 0.81 and 0.19), first label ordering. Only the middle block changes across conditions; the preamble, the assay description, and the elicitation are identical.

FULL.

```
One of the following is responsible for the contamination event. Exactly one is.

    Candidate SIGMA
    Candidate THETA

Before any testing, each candidate is considered equally likely.

5 analysts were each assigned a different assay. Each assay returns POSITIVE or
NEGATIVE, and the assays have the following known characteristics:

    If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
    If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

COMPLETE RESULTS, ALL 5 ASSAYS
    Assay returned POSITIVE
    Assay returned POSITIVE
```

```
    Assay returned NEGATIVE
    Assay returned NEGATIVE
    Assay returned NEGATIVE

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.
```

PARTIAL_BLIND. Neither N nor the filing rule is stated.

```
Analysts were each assigned a different assay. Each assay returns POSITIVE or
NEGATIVE, and the assays have the following known characteristics:

    If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
    If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

REPORTS ON FILE
    Assay returned POSITIVE
    Assay returned POSITIVE
```

PARTIAL_RULED. N and the filing rule are stated; the absences are derivable but not printed.

```
5 analysts were each assigned a different assay. Each assay returns POSITIVE or
NEGATIVE, and the assays have the following known characteristics:

    If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
    If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

Each analyst files a report only when their assay returns POSITIVE.

REPORTS ON FILE
    Assay returned POSITIVE
    Assay returned POSITIVE
```

ARITH_D0. The absences are printed as filing lines.

```
Each analyst files a report only when their assay returns POSITIVE.

COMPLETE FILING RECORD, ALL 5 ANALYSTS
    Assay returned POSITIVE
    Assay returned POSITIVE
    Filed nothing at all
    Filed nothing at all
    Filed nothing at all
Each analyst appears exactly once above.
```

RULED_D30. As PARTIAL_RULED with a stated filing rate.

```
Each analyst files a report only when their assay returns POSITIVE.
Separately from the assay result, each analyst files nothing at all
in 30 percent of cases.

REPORTS ON FILE
    Assay returned POSITIVE
    Assay returned POSITIVE
```

ARITH_D30. As ARITH_D0 with a stated filing rate.

```
Each analyst files a report only when their assay returns POSITIVE.
Separately from the assay result, each analyst files nothing at all
in 30 percent of cases.

COMPLETE FILING RECORD, ALL 5 ANALYSTS
    Assay returned POSITIVE
    Assay returned POSITIVE
    Filed nothing at all
    Filed nothing at all
    Filed nothing at all
Each analyst appears exactly once above.
```

Both label orderings. Every stimulus appears in two orderings; the mirror swaps which printed candidate carries which positive rate. The FULL stimulus above in its second ordering differs only in these two lines, so a fixed label bias is separable from a competent answer:

```
    If SIGMA is responsible, an assay returns POSITIVE with probability 0.19.
    If THETA is responsible, an assay returns POSITIVE with probability 0.81.
```

C Models and inference

Reasoning setting and token budget move together; the "-high" member of a pair raises both. The gate score is mean s_1 on the printed-negatives condition (FULL), the competence gate of Section 3.

alias	model slug	reasoning	max_tokens	temp.	weights	gate s_1
`sol-hi`	`openai/gpt-5.6-sol`	effort high	12000	—	closed	0.200
`sol-no`	`openai/gpt-5.6-sol`	effort none	800	—	closed	0.421
`glm-hi`	`z-ai/glm-5.2`	enabled	16000	0.7	open	0.457
`dsk-hi`	`deepseek/deepseek-v4-flash-0731`	enabled	16000	0.7	open	0.752
`glm`	`z-ai/glm-5.2`	disabled	800	0.7	open	0.764
`deepseek`	`deepseek/deepseek-v4-flash-0731`	disabled	800	0.7	open	1.609
`anchor`	`meta-llama/llama-3.3-70b-instruct`	none	400	0.7	open	1.800

Weight availability is the model provider's; the repository records the slugs above, not licenses (see Appendix D).

The five configurations above the rule are admitted by the gate; `deepseek` and `anchor` are excluded. The `openai/gpt-5.6-sol` model does not accept a temperature, so both `sol` configurations vary by seed alone.

All calls went through the OpenRouter API. Each stimulus was drawn five times, one per seed (seeds 1 through 5); the campaign holds five draws per stimulus. The response was parsed strictly against the schema in Appendix B with no partial credit, and each draw was classified into one of four outcomes: `TRUNCATED` (the completion hit the token budget), `PARSE` (unparseable or off-menu), `REQUEST` (transport or HTTP failure), and `AuthError` (a 401/403, which stops the run immediately). Transient failures were retried up to three times with exponential backoff; authentication failures were not retried. The run resumes by keying on successful draws only, so a failed draw is re-attempted rather than skipped. Token budgets are per configuration, and reasoning tokens are drawn from the same budget, which is why the "-high" configurations carry the larger `max_tokens`.

No claim in Sections 4, 5, or 6 compares two configurations that share a model slug; every comparison is across conditions or across the roster, so the reasoning-and-budget difference within a pair does not enter any reported result.

Draw records carry no wall-clock field, so exact collection dates are not recorded; the only date signal is the modification time of each arm's results file. We describe this in the provenance statement rather than treating those mtimes as collection dates.

D Data and licensing

The repository carries no `LICENSE` file, and per-model terms of use are not recorded in it: it records the model slugs of Appendix C but not their licenses. Model access was through the OpenRouter

API. We state what the repository does and does not contain rather than reproducing terms it does not hold.

E Use of AI assistance

Coding agents and large language models assisted this project in two ways: executing and iterating the analysis pipeline in the repository, including data analysis and figure generation, and drafting and editing the manuscript. Every numeric quantity in the paper was verified independently of that assistance, recomputed from the draw records at the cited commit or, where the quantity is analytic, re-derived from the model's closed form; the design, the claims, and their scoping are the author's.